

Parvez Akhtar

+91 888-232-7732 | parvezakhtar218@gmail.com | [LinkedIn](#) | [GitHub](#) | [Kaggle](#)

PROFESSIONAL SUMMARY

Senior AI/ML Engineer with 5+ years designing and delivering enterprise-scale ML and GenAI systems with a strong focus on reliability, security, and cost. Deep in MLOps and LLMOps: CI/CD pipelines, model-lifecycle automation, RAG (Retrieval-Augmented Generation) workflows, guardrails, and observability of latency, token usage, and cost. Ships models as secure microservices (FastAPI) with RBAC, sets technical standards, and mentors engineers, having cut inference cost ~90% while operationalising AI safely.

TECHNICAL SKILLS

MLOps & LLMOps: CI/CD Pipelines & Automated Testing, Model-Lifecycle Management, CI/CD Eval Gates, Guardrails & Prompt-Injection Defense, Observability (Latency / Token Usage / Cost), Tracing & Eval (Langfuse, OpenTelemetry, RAGAS), Prometheus, SigNoz, ELK Stack, CloudWatch.

GenAI & LLMs: RAG (Hybrid Search, Re-ranking), Embeddings, Vector Indexes (PGVector, Qdrant), LangChain, LlamaIndex, Prompt Engineering, Structured Outputs, Fine-tuning (LoRA/QLoRA, PEFT, Hugging Face), Agentic Workflows (LangGraph, MCP).

APIs & Secure Engineering: Secure Microservices (FastAPI, Spring Boot), RBAC, JWT / Keycloak, Spring Security, Multi-Tenant Isolation.

Cloud & Infra: Azure OpenAI, AWS (SageMaker, Bedrock, Lambda, ECS), GCP, Kubernetes, Docker, Terraform, Temporal.io.

Languages: Python, SQL, Java, TypeScript.

Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, FastAPI, Django.

Data Engineering: Apache Airflow, Snowflake Cortex, Kafka, Elasticsearch, ETL Pipelines, MongoDB, Redis, PostgreSQL.

PROFESSIONAL EXPERIENCE

Bigstep Technologies

Remote

Senior AI/ML Engineer, Lead Architect

04/2025 – Present

- As **Lead Architect**, drove enterprise-scale GenAI delivery across client accounts including **CNBC London** and **VTS**, setting technical standards, mentoring engineers, and translating ambiguous business problems into reliable, secure production architectures.
- **LLMOps & Cost Observability:** Built a **"Strategy Router"** on a **2M+ records/month** extraction engine that classifies each field by complexity and routes ~60% to deterministic parsers, reserving the LLM for the rest, cutting **token spend ~40%** with no measurable accuracy loss while tracking latency, token usage, and cost.
- **GenAI Document Intelligence (VTS):** Built a document-intelligence pipeline abstracting structured terms with **structured-output validation**. Diagnosed a **10-point accuracy regression** and redesigned prompt strategy and image preprocessing to recover the full drop, restoring field-level extraction to its **~88% baseline**.
- **RAG Search & Ranking:** Built a **Hybrid Search** pipeline fusing **SPLADE** (learned sparse) and **Sentence-BERT** (dense) retrieval via **Reciprocal Rank Fusion**, then layered **Cross-Encoder re-ranking** over the top candidates, improving **NDCG@5** by **~15 points (0.71 to 0.86)**.
- **Agentic Automation:** Orchestrated a serverless **multi-agent** framework with **tool / function calling** over **MCP (Model Context Protocol)**. Cut idle compute via a pure **pay-per-execution** model, saving **~\$45k/year**.
- Built an entity-matching pipeline on 22M records using an ensemble of **TF-IDF / Cosine Similarity** and 7 text metrics feeding an **XGBoost classifier (85.3% AUC)**; scaled the platform from MVP to production underpinning a **\$20M+ enterprise deal pipeline**.

Kuttl (Incubated at IIM Lucknow)

Remote

Founding Engineer

12/2023 – 03/2025

- **MLOps lifecycle:** Operationalised the ML lifecycle with **CI/CD evaluation gates (Ragas)** scoring **faithfulness and answer-relevancy**, automatically blocking deployments below threshold and catching quality regressions before production.
- **Fine-tuned Llama-3-8B** (LoRA / PEFT) to replace GPT-4 for structured entity extraction, achieving **98% schema compliance** while reducing per-token inference cost by **90%** and latency by 300ms.
- Engineered a distributed document-parsing pipeline exposed via **FastAPI** microservices with **Temporal.io**, processing 50k+ daily messages with resilient retry logic and dead-letter queues to ensure no data loss.
- Applied semantic caching and request batching for RAG endpoints, improving **P99 latency by 40%** during high-concurrency bursts; built the core backend and AI infrastructure from scratch under strict cost constraints.

Amplicomm

Bengaluru, India

SDE-2

05/2023 – 12/2023

- Established a centralized identity broker using **Keycloak** and **Spring Security**, replacing stateful DB lookups with **stateless JWT** validation to cut login latency by 30%.
- Engineered a **Schema-per-Tenant** architecture with **RBAC (Role-Based Access Control)** for 21 enterprise clients, ensuring data integrity via automated checksum reconciliation pipelines.
- Led a team of 3 through a **Strangler-Fig migration** of a legacy monolith to Spring Boot microservices, enabling independent deploys with instant rollback; enforced **SonarQube** quality gates that cut cyclomatic complexity ~42%.

Amazon

Bengaluru, India

Software Development Engineer (SDE)

03/2021 – 03/2023

- Owned the full SDLC for high-scale microservices, co-developing the SmartPOS billing engine and scaling it to support **3,000+ enterprise merchants** with **99.99% availability**.
- Optimized cloud infrastructure costs by **25% (~\$100k/yr)** via strategic re-architecture of ECS services to use **Spot Instances** and custom auto-scaling policies.
- Developed a data-driven testing framework and **CloudWatch** service-level alarms, identifying **N+1 query bottlenecks** that cut p95 API latency by **20%** and surfaced performance regressions before release.
- Executed the zero-downtime migration of 6 core microservices from GCP to AWS, ensuring data consistency and seamless customer transition.

Publicis Sapient

Remote, India

Associate Data Engineer

09/2020 – 03/2021

- Designed high-frequency data ingestion pipelines using **Apache Airflow** to scrape competitive pricing data at scale, bypassing complex anti-bot protections for competitive intelligence.
- Refined **Elasticsearch** indexing strategies and aggregation queries, reducing query latency by ~40% to enable sub-second dashboard load times for brand performance analytics.

EDUCATION

National Institute of Technology Delhi

Delhi, India

B.Tech in Computer Science & Engineering

2016 – 2020

ACHIEVEMENTS

Amazon "On the Hero Road" Award: Recognized for exceptional impact on team delivery and engineering excellence.

Udacity Computer Vision Nanodegree: Full scholarship recipient (sponsored by Facebook).

Deep Learning Specialization: Completed 5-course specialization (Coursera) with perfect score.